

MODELING SPOT MICROPHONE SIGNALS USING THE SINUSOIDAL PLUS NOISE APPROACH

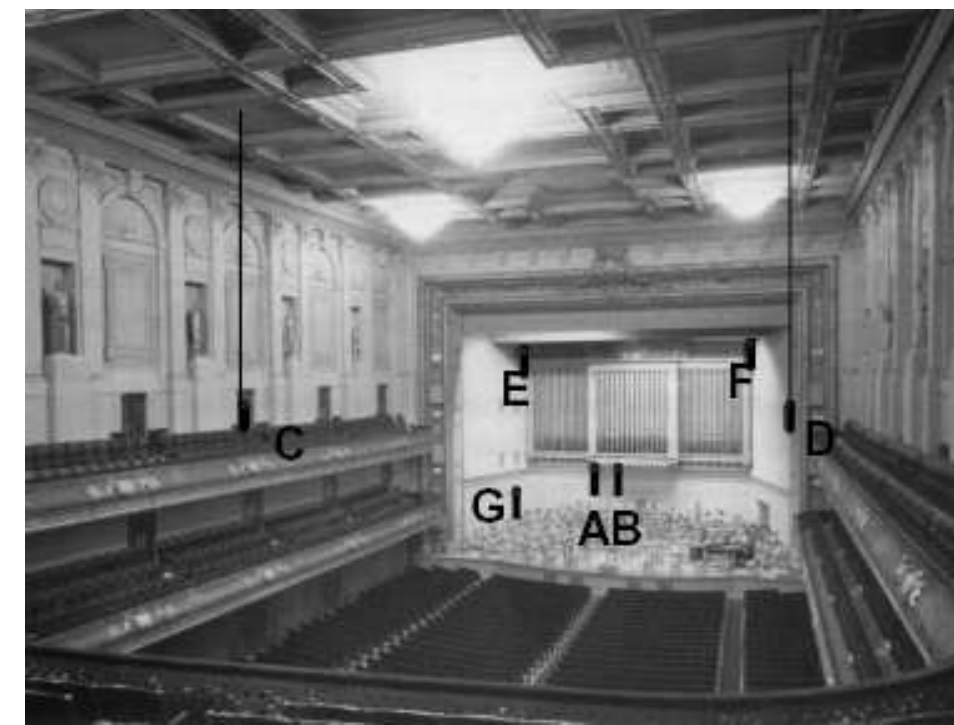
Christos Tzagkarakis, Athanasios Mouchtaris, and Panagiotis Tsakalides
University of Crete & Foundation for Research and Technology - Hellas

Overview

- Description of a method for multichannel audio efficient modeling
 - recreate multiple channels of an existing multichannel audio recording from a smaller set of reference channels (resynthesis)
- Use an enhanced adaptation of the **Sinusoidal plus Noise** model

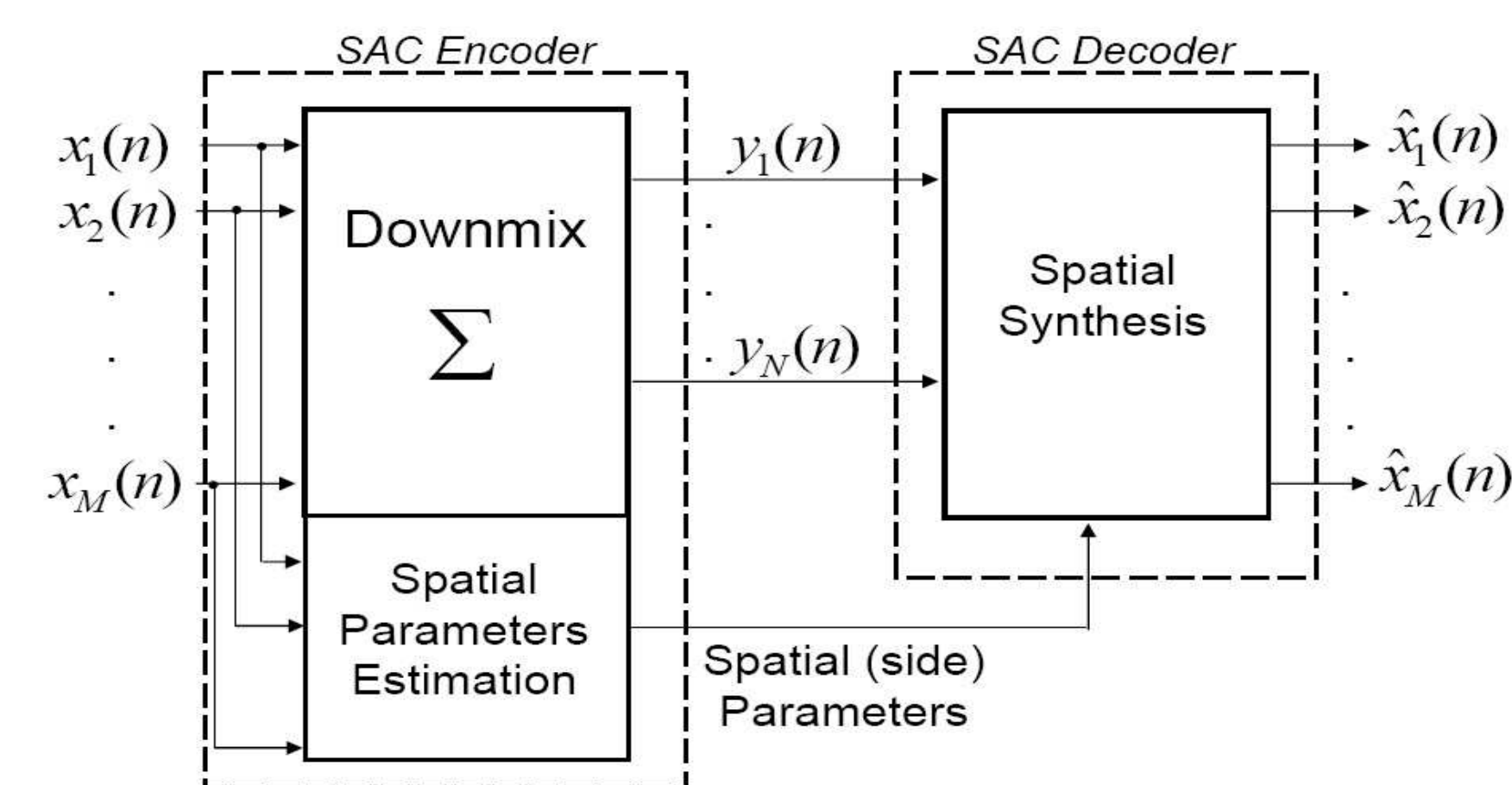
Recording for Multichannel Rendering

- Reverberant microphones:** placed far from the sound source and capture the hall reverberation
- Spot microphones:** placed close to the sound source and capture specific instruments
- A large number of recordings (stem microphone recordings) mixed in order to create the channels of a multichannel recording



Main Idea

- The proposed method focuses on the microphone signals of a performance *before they are mixed*
 - applied to applications such as remote mixing and distributed performances
- The method attempts to model each spot microphone signal with respect to a reference audio channel
 - exploit interchannel similarities*
- The proposed method follows the *Spatial Audio Coding* philosophy



- Employ the Sinusoidal plus Noise approach to model each signal with the sinusoidal parameters (*harmonic part*) and the short-time spectral envelope of the noise (modeling *noise part*)
- For resynthesis apply the **noise transplantation** procedure:
 - noise part for each spot signal is obtained from the noise part of the reference signal by transforming the reference noise part using the noise envelope of each of the remaining signals

Sinusoidal Plus Noise Model

- The *sinusoidal model* represents a signal $s(n)$, harmonic in nature, as a sum of a small number of sinusoids with time-varying amplitudes and frequencies

$$s(n) = \sum_{l=1}^L A_l(n) \cos(\theta_l(n))$$

where $A_l(n)$ and $\theta_l(n)$ is the instantaneous amplitude and phase, respectively, of the l^{th} sinusoid

- Sinusoids cannot be used per se for high-quality audio modeling
- A more accurate representation of audio signals can be achieved with the sinusoidal plus noise model
 - the sinusoids model the *deterministic part* of the signal
 - the rest of the spectral information is left to the sinusoidal noise component $e(n)$

$$s(n) = \sum_{l=1}^L A_l(n) \cos(\theta_l(n)) + e(n)$$

- The noise component $e(n)$ is computed by subtracting the harmonic part from the original signal
- $e(n)$ is modeled using Linear Predictive analysis

$$e(n) = \sum_{i=1}^p \alpha(i) e(n-i) + r_e(n)$$

- $r_e(n)$ is the residual of the noise component
- p is the autoregressive filter order
- the $p+1$ dimensional vector $\vec{\alpha}^T = (1, -\alpha_1, -\alpha_2, \dots, -\alpha_p)$ represents the spectral envelope of $e(n)$

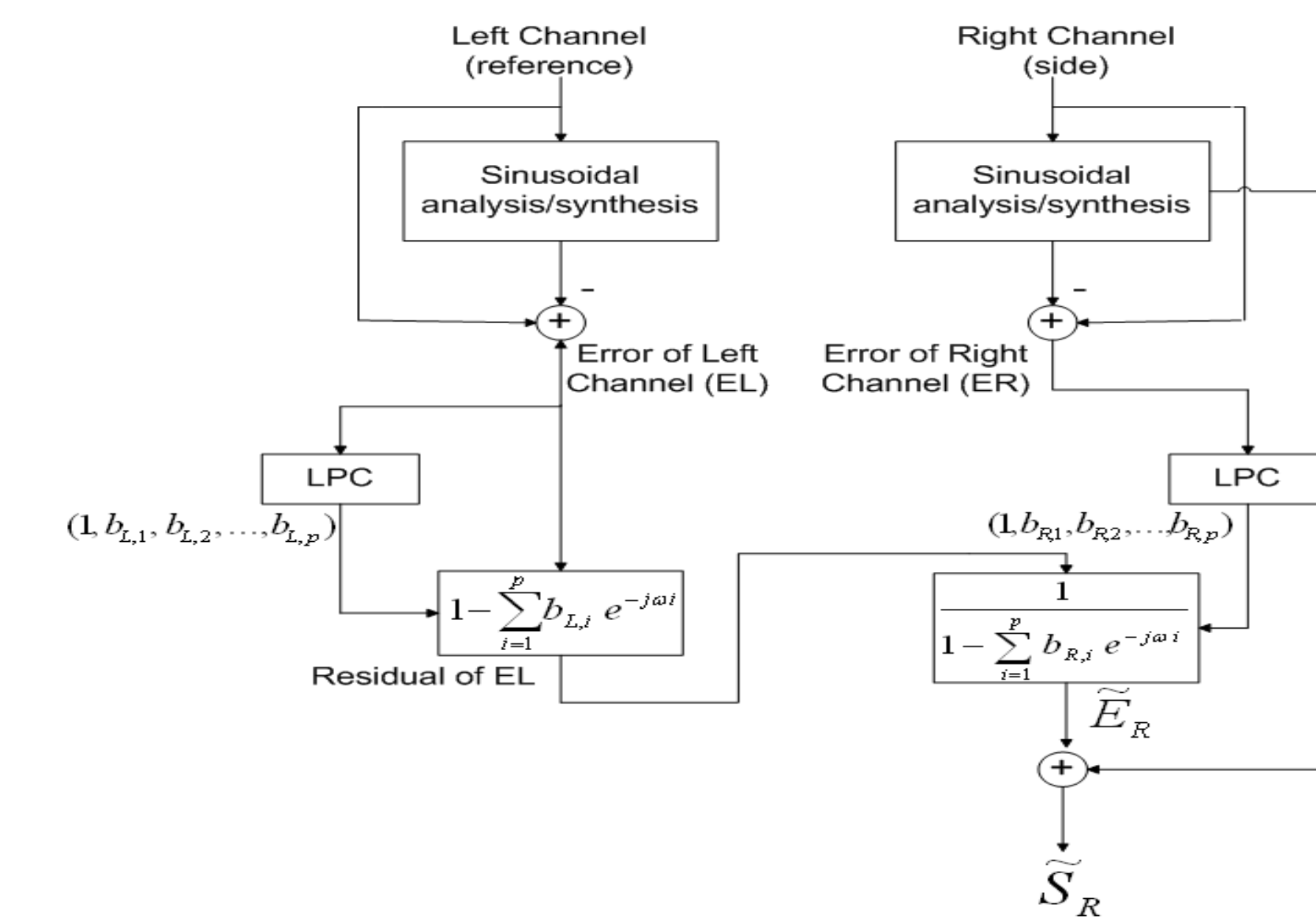
- In the frequency domain the relation above becomes

$$S_e(e^{j\omega}) = \left| \frac{1}{A(e^{j\omega})} \right|^2 S_{r_e}(e^{j\omega})$$

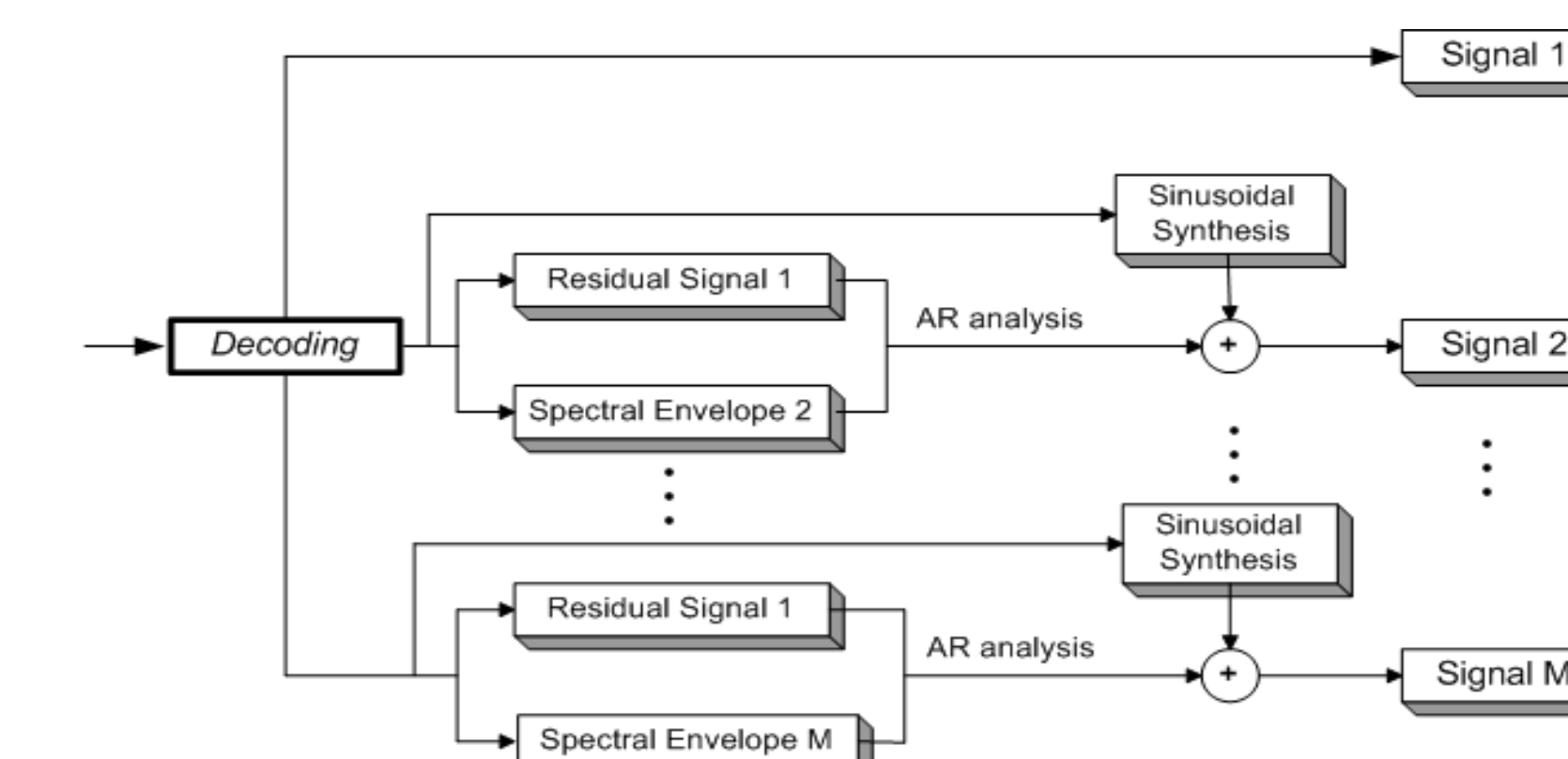
where $S_e(e^{j\omega})$ and $S_{r_e}(e^{j\omega})$ is the power spectrum of $e(n)$ and $r_e(n)$, respectively, and $A(e^{j\omega})$ is the frequency response of the Linear Predictive filter $\vec{\alpha}$

Noise Transplantation

- Consider two spot microphone signals of a music performance of similar acoustical content:
 - left channel $x_L(n)$ (**reference** signal)
 - right channel $x_R(n)$ (**side** signal)



- The side signal is resynthesized using its sinusoidal component and its noise Linear Predictive (LP) filter
 - its sinusoidal component is added to the noise component obtained by filtering with the signal's LP noise shaping filter the LP residual of the reference signal's sinusoidal noise component
- The resynthesis process in the general case of a recording with M spot microphone signals is

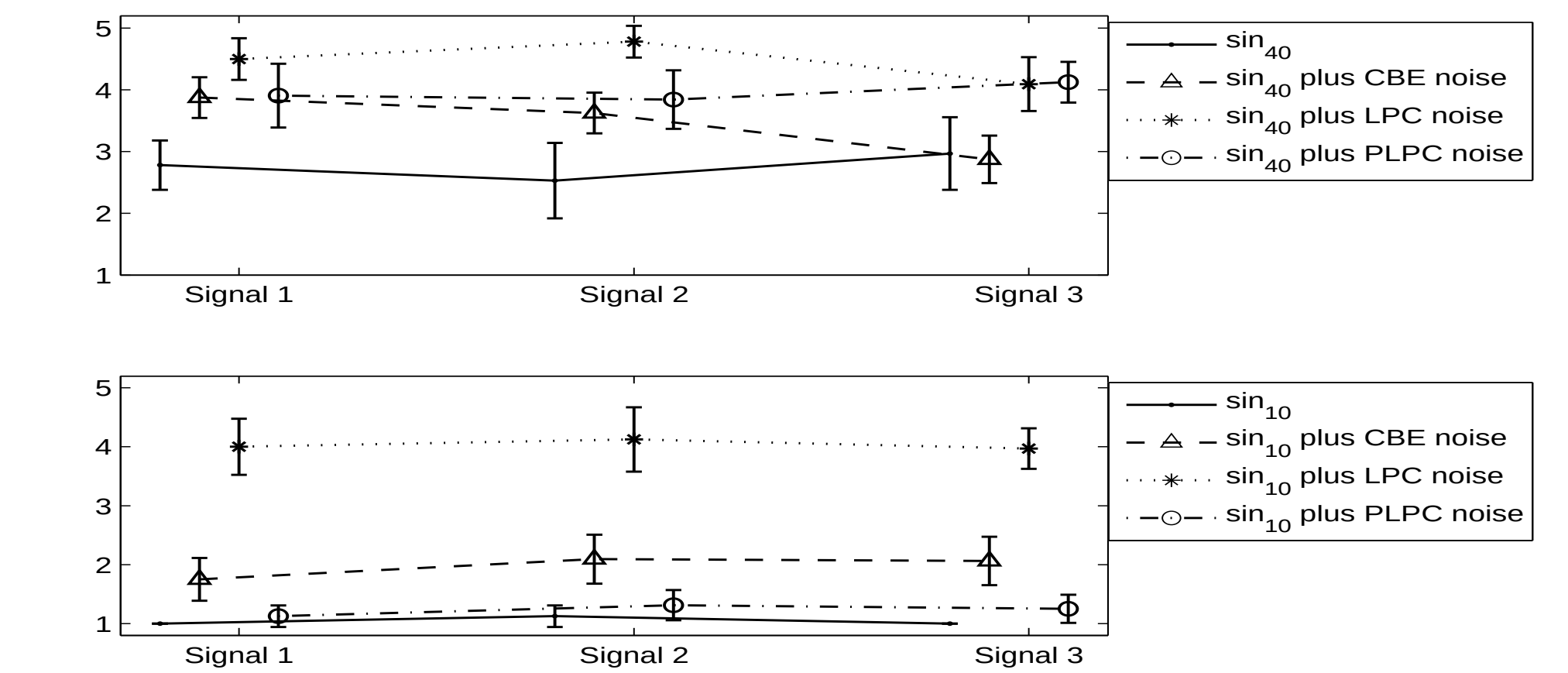


Experimental Results

- Noise transplantation method is compared with:
 - Critical band Energy (CBE)** approach: deals with a noise perception model based on the critical bands of the auditory system
 - Perceptual Linear Predictive Coding (PLPC)** noise shaping model: the residual is modeled by applying LPC in the perceptual domain
- Experimental parameters:**
 - 20 msec. analysis/synthesis frame
 - 50% overlapping (with overlap-add synthesis)
 - LP order for the noise shaping filter is 20
 - total number of sinusoids per frame: 40 and 10

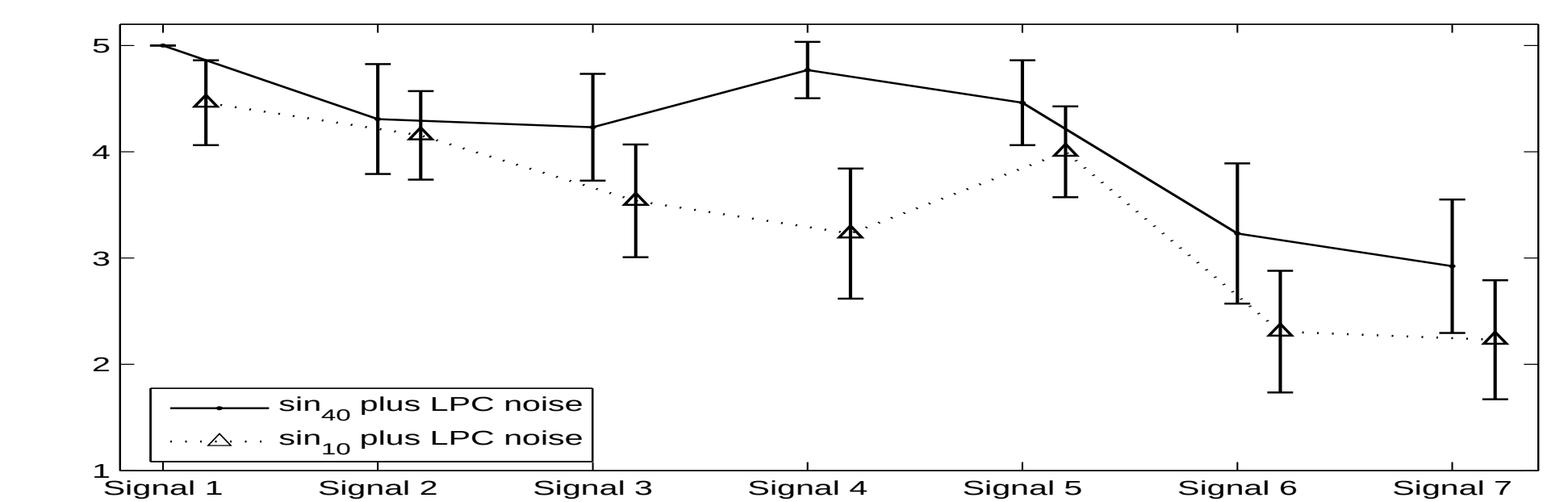
Experiment 1

- Test with two microphone signals of a multichannel recording of a concert hall performance
 - used three 10 sec parts from the two concert hall recordings
- Degradation Category Rating (DCR)** listening tests to compare subjectively the quality of the side signal vs. original
- 16 listeners participated at the DCR test



Experiment 2

- Use seven audio signals to examine the resynthesis quality of the spot signals from a *downmix* sum signal
 - bass plus soprano singer
 - guitar plus rock singer
 - harpischord plus violin
 - female plus male speech
 - trumpet plus violin
 - violin plus guitar
 - violin plus harpsichord
- The instrument referred first is the side signal that is resynthesized from the sum signal (reference)
- 13 volunteers participated at the DCR test



Experiment 3

- ABX test conducted using the seven audio signals listed above, for the case of 40 and 10 sinusoids
 - each listener was presented with the original two instrument recordings used to obtain the sum signal as signals A and B (randomly) and the resynthesized signal (Signal X)
 - associate X with A or B depending on which instrument prevails
- Total ABX score in both cases of 40 and 10 sinusoids $\rightarrow 100\%$
 - no crosstalk is introduced using the proposed method*

Conclusions

- A sinusoidal plus noise model was presented tailored for modeling spot signals of a multichannel recording
- Take advantage of interchannel similarities following SAC philosophy
- Allows for remote applications based on the availability of the spot signals at the decoder side
- The perceived quality does not change by decreasing the number of the sinusoidal parameters
 - low bitrate